Product Brief

Funnel Drop-off Diagnosis Tool

Track 3 — Marketing & Growth

From Workflow Pain to a Working AI Product

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Deliverable: 1 of 2 — Product Brief (PDF)

SECTION 1

Problem Statement

When a funnel shows a sudden drop in conversion, growth managers at D2C startups don't have a fast, reliable way to diagnose *why*. Their current workflow is fragmented — they bounce between analytics dashboards like GA4 or Mixpanel to spot the drop, then wait 2–3 days for a data analyst to pull deeper reports, and in the meantime make decisions based on gut feel and past experience.

By the time a hypothesis reaches testing, days or weeks have passed. The analysis itself is often shallow — dashboards show *where* users drop, but not *why*, and certainly not *what to test next*. This gap between seeing a problem and acting on it costs real revenue, especially in fast-moving D2C environments where campaign cycles are short and CAC is rising.

SECTION 2

User Persona

Attribute	Detail
Name	Neha Sharma
Role	Growth Product Manager at a D2C personal care brand (Series A, 40-person team)
Tools	Google Analytics 4, Mixpanel, Google Sheets, Slack, Meta Ads Manager
Day-to-day	Owns the purchase funnel from landing page to checkout. Reports weekly on conversion metrics. Runs 2–3 A/B tests per month.
Frustrations	Spends Monday mornings manually pulling numbers across tools. When a drop happens, she can see which step broke but not why. The one data analyst on the team is backlogged. She ends up guessing what to test.
Goal	Diagnose funnel drops in minutes, not days — and walk away with a concrete experiment to run.

SECTION 3

JTBD Statement

> *"When I see a sudden drop in my funnel conversion, I want to quickly understand what's causing it and get testable hypotheses, so I can run the right experiment this week instead of guessing for the next two."*

Context of use:

- Trigger: Weekly metrics review reveals an unexpected conversion dip
- Frequency: 2–4 times per month
- Urgency: High — each day of undiagnosed drop erodes revenue
- Current alternative: Manual dashboard triage + analyst queue (2–3 day turnaround)

SECTION 4

Why AI? (Critical Thinking)

What logic alone handles (no AI needed):

- Calculating step-over-step conversion rates from raw input
- Flagging abnormal drops using statistical thresholds (e.g., >1.5 standard deviations below the funnel's average conversion)
- Assigning severity tags — Normal, Warning, Critical
- This is deterministic math. It is reliable and reproducible.

What logic alone cannot handle:

- Interpreting why a specific step is dropping. A 38% drop at "Add to Cart → Checkout" means something very different from a 38% drop at "Landing Page → Product Page." Step descriptions carry context that rules can't parse.
- Generating hypotheses that account for common UX, pricing, trust, and friction patterns across e-commerce funnels.
- Suggesting a specific experiment design — what to change, what to measure, how long to run it.

Why AI fits here:

The LLM acts as a pattern-matching layer on top of structured logic output. It does not detect the drop — the logic layer does that. It *interprets* the drop using step context and generates actionable next steps. This is a reasoning task over semi-structured data, which is exactly where LLMs add value without hallucination risk on the numbers themselves.

SECTION 5

Feature List — Value vs. Effort

Feature	Value	Effort	Priority
Structured funnel input form	High	Low	P0
Auto-detection of abnormal drops (logic layer)	High	Medium	P0
AI-generated drop-off reasons	High	Medium	P0

Feature	Value	Effort	Priority
Testable hypotheses output	High	Medium	P0
Experiment suggestion	High	Medium	P0
Confidence score per analysis	Medium	Low	P0
Past analysis storage (DB)	Medium	Medium	P1
Admin dashboard with history	Medium	Medium	P1
Error handling + validation	Medium	Low	P0
CSV upload for funnel data	Low	Medium	P2 (stretch)
Multi-funnel comparison	Low	High	P2 (stretch)

P0 features form the MVP. P2 items are stretch goals if time permits.

SECTION 6

Failure Mode Analysis

Failure Mode	Why It Happens	Mitigation
Generic advice ("improve UX")	Prompt lacks specific step context or severity data	Logic layer feeds flagged steps + descriptions + severity into prompt — forces specificity
Hallucinated metrics	LLM invents numbers not in the input	AI never touches raw data. All numbers come from logic layer. Prompt says: do not generate statistics
Irrelevant hypotheses	LLM doesn't understand the funnel domain	Prompt includes funnel category (D2C, SaaS) and step descriptions as mandatory context
False confidence	Every analysis looks equally confident regardless of data quality	Confidence score calculated before AI, based on data completeness and severity of flags
API failure / timeout	OpenAI rate limit or downtime	Graceful error state. Logic layer output still displays even if AI call fails
Garbage input data	Conversion rates >100%, negative users, empty steps	Input validation layer catches these before anything runs

SECTION 7

AI Trade-offs

Accuracy vs. Speed

GPT-4o gives better hypothesis quality but costs more and is slower (~3–5 s). GPT-3.5-turbo is faster and cheaper but outputs more generic suggestions. Starting with GPT-4o-mini as a balanced middle ground.

Cost vs. Quality

Each analysis uses ~800–1,200 tokens. At GPT-4o-mini pricing that's roughly Rs. 0.5–1 per analysis — acceptable for a tool used 5–15 times per week.

Determinism vs. Creativity

Temperature set to 0.3 — low enough to avoid wild suggestions, high enough to produce varied hypotheses across similar inputs.

Dependency Risk

If the OpenAI API goes down, the logic layer still works. The product degrades gracefully — users see flagged drops and severity, just without AI-generated hypotheses.

SECTION 8

Cost & Sustainability

Usage Tier	Analyses / Month	Token Cost (est.)	Infra Cost	Total
Light	100	~Rs. 75	Rs. 0 (free tier)	~Rs. 75/mo
Medium	500	~Rs. 375	Rs. 0 (free tier)	~Rs. 375/mo
Heavy	1,000	~Rs. 750	~Rs. 800 (paid tier)	~Rs. 1,550/mo

The logic layer handles the heavy lifting. AI is called once per analysis, not in a loop. Token usage is predictable because input is structured (not free-form chat). This keeps costs linear and controllable.

SECTION 9

Reflection

The biggest shift in thinking was realizing that the AI isn't the product — the *logic layer* is. The statistical detection of abnormal drops is what makes this tool trustworthy. AI just makes it actionable.

I initially wanted to let users paste raw text and let AI figure it out. Forcing structured input felt limiting at first, but it's exactly what makes the output reliable and the failure modes manageable.

Designing failure modes *before* building changed how I approached prompt design. I wasn't trying to make AI sound smart — I was trying to prevent it from saying something useless.

If I had more time, I'd add funnel comparison across time periods and a feedback loop where users rate hypothesis quality to improve future prompt construction.